



SAFETY DATA SHEET

1. Identification

Product Identifier	Premier 1.3
Other means of Identification	Sodium Hypochlorite Solution, Bleach Solution, Bleach Liquor, Hypo-solution, Bleach, Liquid Bleach Cleaner
Recommended Use	None known
Recommended Restrictions	None known
Company/Undertaking Identification	Premier Water & Energy Technology, Inc. 11481 Columbia Park Drive, West Jacksonville, FL 32258
Telephone, Fax	904-268-1152, 904-268-6851
Emergency Telephone	800-255-3924, Chemtel, Inc.

2. Hazard(s) Identification

Physical Hazards	Not classified	
Health Hazards	Skin corrosion/irritation	Category 1B/2
	Serious eye damage/eye irritation	Category 2
	Aquatic Acute	Category 1
Environmental Hazards	Not classified	
OSHA Defined Hazards	Not classified	

Label Elements



Signal Word

Danger

Hazard Statement

Causes severe skin burns and eye damage. Causes serious eye irritation. Very toxic to aquatic life.

Precautionary Statement

Prevention	Do not breathe dusts or mists. Wash hands or any exposed skin areas thoroughly after handling. Avoid release to the environment. Wear protective gloves/protective clothing/eye protection/face protection.
Response	If swallowed: Rinse mouth. Do NOT induce vomiting. If on skin (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower. If inhaled: Remove person to fresh air and keep comfortable for breathing. If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a poison center/doctor. Wash contaminated clothing before reuse.
Storage	Store locked up.
Disposal	Dispose of contents/container in accordance with local/regional/national/international regulations.

Hazard(s) Not Otherwise Classified (HNOC)

Do not mix with acids or other compatible materials.
Contact with acids liberates toxic gas.

Supplemental Information

This material and its container must be disposed of as hazardous waste.

3. Composition/Information on Ingredients
Mixtures

Chemical Name	Common Name and Synonyms	CAS Number	%
Sodium Hypochlorite		7681-52-9	10.0-20.0
Sodium Hydroxide		1310-73-2	0.1-0.4
Water		7732-18-5	79.7-89.9

*Designates that a specific chemical identity and/or percentage of composition has been withheld as a trade secret.

4. First-aid Measures
Inhalation

If product vapors or mists cause respiratory irritation or distress, move the exposed person to fresh air immediately. If breathing is difficult or irregular, administer oxygen; if respiratory arrest occurs, start artificial respiration by trained personnel. Loosen tight clothing such as a collar, tie, belt or waistband. If symptoms persist, seek medical attention immediately.

Skin contact

Flush skin with large amounts of water while removing contaminated clothing. Wash affected area with soap and water. Wash contaminated clothing and shoes thoroughly before reuse. Seek prompt medical attention if rash develops.

Eye contact

Immediately flush eyes with large amounts of water for 15 minutes, occasionally lifting upper and lower lids. Remove contact lenses after the first 5 minutes and continue washing. Obtain immediate medical attention, preferably from an ophthalmologist.

Ingestion

Rinse mouth with water. Do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Get medical attention immediately.

Most important symptoms/effects, acute and delayed

Eyes: Causes severe eye irritation and burns. Symptoms include redness, pain, itching, burning sensation and tearing. Material is extremely destructive to eyes, mucous membranes and surrounding tissues.

Skin: Causes severe skin irritation and burns. Symptoms include redness, pain, itching and burning sensation. May be harmful if absorbed through the skin.

Inhalation: Vapors and mists may be harmful if inhaled, causing sore throat and cough. Material is extremely destructive to the tissue of the mucous membranes and upper respiratory tract.

Ingestion: May cause severe gastrointestinal tract irritation with abdominal pain, burning sensation, cough, diarrhea, sore throat and vomiting. May cause burns and irritation to mucous membranes of the mouth and to tissues of the digestive tract.

Chronic: Repeated or prolonged contact with spray mist may produce chronic eye irritation, severe skin irritation and/or respiratory tract irritation leading to frequent attacks of bronchial infection.

5. Fire-fighting Measures
Suitable extinguishing media

Material does not burn. Use fire extinguishing media appropriate for surrounding materials.

Unsuitable extinguishing media

None listed

Specific hazards arising from the chemical

Closed containers may explode (due to the build-up of pressure) when exposed to extreme heat. During emergency conditions overexposure to toxic decomposition products may cause a health hazard. Fire may cause the evolution of chlorine, hydrogen chloride gas and chlorine oxides. Symptoms may not be immediately apparent. Obtain immediate medical attention.

Special protective equipment and precautions for firefighters

Full protective equipment including self-contained breathing apparatus should be used. Water may be used to cool closed containers to prevent pressure build-up and possible autoignition or explosion when exposed to extreme heat. If possible, firefighters should control run-off water to prevent environmental contamination.

6. Accidental Release Measures

Personal precautions, protective equipment, and emergency procedures Methods and materials for containment and cleaning up

Avoid breathing vapors/mists. Avoid contact with skin and eyes. Wear appropriate protective clothing designated in Section 8. Ventilate the area. Evacuate personnel to safe areas.
Cover drains. Cover with a large quantity of inert absorbent (e.g. sand, vermiculite, kitty litter, dry earth). Do not use combustible materials such as saw dust. Collect product using a shovel and place into approved container for proper disposal as hazardous waste. For large spills use water spray to divert vapor drift. Observe possible material restrictions (section 7.2 and 10.5). Clean contaminated area with water. Do not mix with other cleaning agents that may liberate chlorine gas vapors.

US Regulations (CERCLA) require reporting spills and releases to soil water and air in excess of reportable quantities. Reportable quantity (RQ) for hypochlorite solutions is 45.36 kg (100 lbs).

Reportable Quantity (RQ): 100 lbs or 45.36 kg (approximately 100 gal or 378.5 L of Premier 1.3. In the event of a spill (e.g. defined as any release to the environment), call Premier Water & Energy Technology, Inc. and/or the emergency contact numbers as soon as possible for assistance.

Environmental precautions

Avoid dispersal of spilled material or run-off and prevent contact with soil and entry into drains, sewers or waterways. Contain and recover liquid when possible.

*****For releases higher than the Reportable Quantity (RQ), you must notify the State Emergency Response Commission at (800) 320-0519 AND the National Response Center at (800) 424-8802 or (202) 267-2675 within 15 minutes!!!*****

Plan in advance for an occupational release and have necessary equipment and neutralization agents on-site.

7. Handling and Storage

Precautions for safe handling

Observe label precautions. Avoid contact with skin and eyes. Wear all appropriate protective equipment specified in Section 8. Wash thoroughly after handling. Keep containers closed when not in use. Use proper equipment for lifting and transporting all containers.

Advice on protection against fire and explosion

Material is non-flammable and non-combustible.

Conditions for safe storage, including any incompatibilities

Keep in cool, dry, ventilated storage areas in closed containers. Protect against physical damage. Isolate from incompatible substances. Do not store near acids, heat, oxidizable materials or organics.

Store in a receptacle equipped with a vent. Transfer only to approved containers having correct labeling. Containers that have been opened should be carefully resealed and kept upright to prevent leakage. Do not take internally. Keep locked up and out of reach of children.

8. Exposure Controls/Personal Protection

Occupational exposure limits

US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Components	Type	Value
Sodium Hypochlorite (CAS 7681-52-9)	TWA	2 mg/m3 (skin)

Sodium Hydroxide (CAS 1310-73-2)	TWA	2 mg/m3
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US. ACGIH Threshold Limit Values

Components	Type	Value
Sodium Hypochlorite (CAS 7681-52-9)	TWA	0.5 ppm as CL2
	STEL, A4	1 ppm as CL2
Sodium Hydroxide (CAS 1310-73-2)	Ceiling	2 mg/m3

AIHA (WEEL)

Components	Type	Value
Sodium Hypochlorite (CAS 7681-52-9)	STEL	2 mg/m3

Appropriate engineering controls

Technical measures and appropriate working operations should be given priority over the use of personal protective equipment. Use adequate ventilation. Local exhaust is preferable. See section 7.1.

Individual protection measures

Wear protective clothing to prevent repeated or prolonged contact with product. Protective clothing needs to be selected specifically for the workplace, depending on concentrations and quantities of hazardous substances handled. The chemical resistance of the protective equipment should be enquired at the representative supplier.

Eye/Face Protection

Wear tightly fitting protective goggles and a face shield (8-inch minimum). Refer to 29 CFR 1910.133, ANSI Z87.1 or European Standard EN 166.

**Skin Protection/
Hand Protection**

Wear gloves recommended by glove supplier for protection against materials in section 3. Gloves must be inspected prior to use. Gloves should be impermeable to chemicals and oil. Breakthrough time of selected gloves must be greater than the intended use period. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product.

Other

Wear impervious, protective chemical resistant clothing including boots, gloves, lab coat, apron or coveralls as appropriate to the situation to prevent skin contact.

Respiratory Protection

Always use an approved respirator when vapor/aerosols are generated. Where risk assessment shows air-purifying respirators are appropriate use a full-faced respirator with multi-purpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

Environmental Exposure Controls

Do not empty into drains.

General Hygiene Considerations

Facilities storing or using this material should be equipped with an eyewash station and safety shower. Change contaminated clothing. Preventive skin protection is recommended. Wash hands thoroughly after use, before eating, drinking or using the lavatory and at the end of the workday.

9. Physical and Chemical Properties

Appearance
Color

Clear, greenish yellow

Physical State

Liquid

Odor

Pungent, chlorine-like

Odor Threshold

Not available

Molecular Weight

74.44 (sodium hypochlorite)

Chemical Formula

NaOCl

pH

11-13

Freezing Point

-13.9° C (7° F)

Initial Boiling Point and Boiling Range	100° C (212° F) – lowest known value
Flash Point	Not available
Evaporation Rate	<1 (BuAc = 1)
Flammability (solid, gas)	Not applicable
Upper/Lower Flammability or Explosive limits	
Flammability Limit – lower (%)	Not available
Flammability Limit – upper (%)	Not available
Explosive Limit – lower (%)	Not available
Explosive Limit – upper (%)	Not available
Vapor Pressure	Not available
Vapor Pressure Temp.	Not available
Vapor Density	Not available
Relative Density	1.15 – 1.17 g/ml (9.597 – 9.764 lb/gal)@ 60° F
Relative Density Temperature	Not available
Solubility(ies)	
Solubility (water)	Complete
Partition Coefficient (n-octanol/water)	Not available
Auto-ignition Temperature	Not available
Decomposition Temperature	110° C (230° F)
Viscosity	Not available
Viscosity Temperature	Not available
Other Information	
Percent Volatile	Not available
Density	Not available
Specific Gravity	Not available
Explosive properties	Not available
Flammability class	Not available

10. Stability and Reactivity

Reactivity	Slowly decomposes on contact with air. Rate increases with the concentration and temperature. Exposure to sunlight accelerates decomposition.
Chemical stability	Stable under recommended storage conditions. Slowly decomposes on contact with air. Rate increases with the concentration and temperature. Exposure to sunlight accelerates decomposition. Sodium hypochlorite becomes less toxic with age.
Possibility of hazardous reactions	Avoid excessive heat and sources of ignition. Flammable hydrogen may be generated from contact with metals such as: aluminum, brass, tin, zinc and alloys of these metals. Avoid contact with acids, halogenated organics, organic nitro compounds and glycols. Hazardous gases may be generated from contact with acids, ammonium hydroxide (aqua ammonia) or cleaners containing ammonia compounds. Violent reactions may occur with some organic compounds. Sodium hypochlorite reacts readily with various reducing sugars (e.g. fructose, galactose, maltose, dry whey solids) to produce carbon monoxide
Conditions to avoid	Precautions should be taken including atmospheric monitoring of the tank to ensure safety of personnel. Hazardous polymerization will not occur.
Incompatible materials	Light, heat, air and contact with incompatible materials Ammonia, amines, ammonium salts, aziridine, methanol, phenyl acetonitrile, cellulose, ethyleneimine, organic materials, oxidizable metals/powdered metals, acids, soaps and bisulfates. Forms shock-sensitive mixtures with certain other materials.
Hazardous decomposition products	Thermal decomposition products include chlorine gas, hydrogen chloride gas, hydrochloric acid, sodium oxide. Decomposition rate increases with temperature.

11. Toxicological Information

Information on likely routes of exposure

Inhalation	May cause severe bronchial irritation, sore throat with possible blistering, coughing, stomatitis, nausea, labored breathing, shortness of breath and pulmonary edema. 10-20 mg/m ³ causes burning of the nose and throat; 40-60 mg/m ³ may be fatal. If sufficient amounts are absorbed, may cause effects as detailed in acute ingestion.
Skin contact	Extent of damage depends on concentration, pH, and volume of solution and duration of contact. May cause redness, pain, blistering, itchy eczema and chemical burns. Sensitization reactions are possible in previously exposed persons. Skin irritation - 24 h (Rabbit)
Eye contact	Rabbit, Adult – 10 mg, moderate irritation May cause redness, pain, and blurred vision. Solutions of 5% splashed in human eyes have caused a burning sensation and later only slight superficial disturbance of the corneal epithelium which cleared completely in the next day or two without special treatment. However, one animal study reports a 5% solution causing only moderate irritation with clearing within 7 days. A higher concentration of 15% tested on rabbit eyes caused immediate severe pain, hemorrhages, rapid onset of ground-glass appearance of the corneal epithelium, moderate bluish edema of the whole cornea, chemosis and discharge for several days. Such eyes have sometimes healed in 2-3 weeks with slight or no residual corneal damage, but they had neovascularization of the conjunctiva and distortion of the nictitating membrane by scarring.
Ingestion	May cause irritation and erosion of the mucous membranes, vomiting (possibly bloody) and abdominal pain and spasms. A drop in blood pressure, shallow respiration, edema (possibly severe) of pharynx, larynx, and glottis, confusion, convulsions, delirium and coma may occur. Cyanosis and circulatory collapse are possible. Esophageal or gastric perforation and strictures are rare. Death may occur, usually due to complications of severe local injury such as toxemia, shock, perforations, hemorrhage, infection and obstruction. Massive ingestions may produce fatal hyperchloremic metabolic acidosis or aspiration pneumonitis.
Further Data	Handle in accordance with good industrial hygiene and safety practice.

Information on toxicological effects

Acute toxicity	Harmful if swallowed.
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Components	Species	Test Results
SODIUM HYPOCHLORITE (CAS 7681-52-9)		
Acute		
Oral	TDLo	Woman
	TDLo	Man
	LD50	Mouse
	LD50	Rat
		1 gm/kg
		45 mg/kg intravenous
		5,800 mg/kg
		140 mg/kg – 9 weeks

*Estimates for product may be based on additional component data not shown.

Respiratory or skin sensitization

Respiratory sensitization	Not a respiratory sensitizer.
Skin sensitization	May cause allergic skin reaction
Germ cell mutagenicity	Mutation in microorganisms – Salmonella typhimurium 1mg / plate (-S9) DNA repair – Escherichia coli 20 µg/ disc; DNA damage – Escherichia coli 420 µmol/L;

Phage inhibition capacity – *E. coli* 103 µg/ well
 Micronucleus test - non-mammalian species multiple 200 ppb
 Cytogenetic analysis - non-mammalian species multiple 120 µg/L
 Cytogenetic analysis – human lymphocyte 100 ppm 24hour(s)
 Sister chromatid exchange – human embryo 149 mg/ L
 Cytogenetic analysis – hamster lung 100 mg/ L

OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050)

Not Listed

Reproductive Toxicity	This product is not expected to cause reproductive or developmental effects.
Specific Target Organ Toxicity – single exposure	No data available
Specific Target Organ Toxicity – repeated exposure	May cause allergic skin reactions, dermatitis (allergic and contact) and asthma or bronchitis. Sensitization reactions are reported in individuals who are exposed in small amounts through their water supply. High doses have caused sperm abnormality in mice.
Aspiration Hazard	No test data available. Risk of serious damage to lungs by aspiration.
Chronic Effects	Persons with impaired respiratory function may be more susceptible to the effects of this substance.

Sodium Hypochlorite (hypochlorite salts) is listed by IARC as a Group 3 Carcinogen – Not classifiable as to its carcinogenicity to humans. Sodium Hydroxide is not listed by IARC. None of the components of this product are listed as carcinogens by ACGIH, IARC, NTP or OSHA. No data is available regarding its mutagenicity and/or teratogenicity of this material, nor is there any available data that indicates it causes adverse developmental and/or fertility effects.

12. Ecological Information

Ecotoxicity This product is very toxic to aquatic organisms.

Components	Species	Test Results
Aquatic		
Acute and Prolonged	LC50	Pimephales promelas (Fathead minnow) 96 h: 0.22 – 0.62 mg/L Oncorhynchus clarki (Cutthroat trout) 96 h: 0.94 µg/L (mortality)
Acute	EC50	Daphnia magna (Water flea), 96 h: 2.1 mg/L
Acute	LC50	Protozoan phylum (Protozoa), 7 h: 31.6 µg/L
Acute	LC50	Algae, phytoplankton, algal mat (Algae), 96 h: 90 µg/L (mortality)
Acute	EC50	Desmodesmus subspicatus (Green algae), 24 h: 28 mg/L
Acute	Biomass reduction	Potamogeton crispus (Curled pond weed), 35h: 23 µg/L
Acute	Chlorophyll Threshold	Aquatic community, 28 d: 2.1 µg/L

*Estimates for product may be based on additional component data not shown.

Persistence and degradability	The methods for determining the biological degradability are not applicable to inorganic substances.
Bioaccumulative Potential	Data not available. Bioaccumulation is not expected
Mobility in Soil	Product is mobile in water.
Other Adverse Effects	This material is a very toxic to aquatic life. Do not allow material to run into surface waters, wastewater or soil.

13. Disposal Considerations

**Waste Treatment
Methods**

The generation of waste should be avoided or minimized whenever possible. This material is subject to disposal regulations under U.S. EPA 40 CFR Parts 261 and 262. Container should be disposed of in a safe way as empty containers may contain product residue. Leave chemicals in original containers. No mixing with other waste. Handle unclean containers like the product itself. Incinerate in an approved facility. Do not incinerate closed container. Dispose of in accordance with the Directive 2008/98/EC as well as other national, federal, state/provincial and local laws and regulations.

No waste code according to the European Waste Catalogue can be assigned for this product, as the intended use dictates the assignment. The waste code is established in consultation with the regional waste disposer.

14. Transport Information
DOT

UN number	UN1791
UN Proper Shipping Name	Hypochlorite Solutions
Transport Hazard Class(es)	
Class	8
Subsidiary Risk	-
Packing Group	III
Special Precautions per User	Read safety instructions, SDS and emergency procedures before handling.
ERG Number	157
DOT information on packaging may be different from that listed.	

IMO/IMDG (Water Transportation)

UN number	UN1791
UN Proper Shipping Name	Hypochlorite Solutions
Transport Hazard Class(es)	
Class	8
Packing Group	III
Marine Pollutant	No
EMS Number	F-A, S-B

ICAO/IATA (Air Transportation)

UN number	UN1791
UN Proper Shipping Name	Hypochlorite Solutions
Transport Hazard Class(es)	
Class	8
Packing Group	III
Quantity Limitations	49 CFR 175075 – Cargo Aircraft Only: 60L, Passenger Aircraft: 5L

RID/ADR (Rail Transportation)

UN number	UN1791
UN Proper Shipping Name	Hypochlorite Solutions
Transport Hazard Class(es)	
Class	8
Packing Group	III

Hazard Symbols


15. Regulatory Information

US Federal Regulations

OSHA Hazard Communication Standard	This product is a "Hazardous Chemical" as defined by the OSHA Hazard Communication Standard, 29 CFR 1910.1200. CORROSIVE
OSHA PSM	Not regulated under OSHA Process Safety Management Standard (PSM) 29 CFR 1910.119
EPA RMP	Not regulated under EPA Risk Management Standard (RMP) 40 CFR Part 68
EPA FIFRA	This product is a registered Pesticide under the Federal insecticide, Fungicide and Rodenticide Act (FIFRA) 40 CFR Part 150

TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)

All components of this product are listed on the Toxic Substance Control Act (TSCA) Inventory. This product not subject to TSCA 12(b) Export Notification.

CERCLA Hazardous Substance List (40 CFR 302.4)

Sodium Hypochlorite (CAS 7681-52-9)	RQ – 45.36 kg (100 lbs)
Sodium Hydroxide (CAS 1310-73-2)	RQ – 453.59 kg (1,000 lbs)

Superfund Amendments and Reauthorization Act of 1986 (SARA)

SARA 311/312 Hazard Categories	This product is subject to the reporting requirements of Section 311/312 of the Emergency Planning and Community Right-to Know Act of 1986.
Acute -	Yes
Chronic -	No
Fire -	No
Reactive -	No
SARA 302/304 Extremely hazardous substance	No components of the product exceed the threshold (de minimis) reporting levels established by of these sections of Title III of SARA.
SARA 302/304 Emergency Planning & Notification	No components of the product exceed the threshold (de minimis) reporting levels established by of these sections of Title III of SARA.

SARA 313 (TRI reporting)

None of the chemicals in this product exceed the threshold (de minimis) reporting levels established by Section 313 of the Emergency Planning and Community Right-to Know Act of 1986.

Other federal regulations

Clean Air Act (CAA)

This product does not contain any chemicals that are listed as Hazardous Air Pollutants (HAPs) designated in CAA Section 112 (b).

This product does not contain any Class 1 Ozone depleters.

This product does not contain any Class 2 Ozone depleters.

Clean Water Act

Sodium hypochlorite, sodium hydroxide and hypochlorite solutions are listed as Hazardous Substances under the CWA.

None of the chemicals in this product are listed as Priority Pollutants under the CWA.

None of the chemicals in this product are listed as Toxic Pollutants under the CWA.

US State Regulations

US. California Proposition 65

California Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65): This material is not known to contain any chemicals currently listed as carcinogens or reproductive toxins.

Other U.S. State Inventories

Sodium hypochlorite (CAS #7681-52-9) is found on the following State Hazardous Substance Inventories and/or Right-to-Know lists: CA, DE, MA, MN, NY, NJ, PA.

Sodium hydroxide (CAS #1310-73-2) is found on the following State Hazardous Substance Inventories and/or Right-to-Know lists: CA, DE, ID, MA, MN, NY, NJ, PA, WA, WI .

European Economic Community

WGK, Germany (Water danger/protection): 2

International Inventories

Country(s) or region	Inventory name	On inventory (yes/no)*
Australia	Australian Inventory of Chemical Substances (AICS)	Yes
Canada	Domestic Substances List (DSL)	Yes
Canada	Non-Domestic Substances List (ND SL)	Yes
China	Inventory of Existing Chemical Substances in China (I ECSC)	Yes
Europe	European Inventory of Existing Commercial Chemical Substances (EINECS)	Yes
Japan	Inventory of Existing and New Chemical Substances (ENCS)	Yes
Korea	Existing Chemicals List (ECL)	Yes
New Zealand	New Zealand Inventory	Yes
Philippines	Philippine Inventory of Chemicals and Chemical Substances (PICCS)	Yes
United States & Puerto Rico	Toxic Substances Control Act (TSCA) Inventory	Yes

*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s). A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

16. Other Information, including date of preparation or last revision
Hazardous Material Information System (HMIS)

HEALTH	2
FLAMMABILITY	0
REACTIVITY	1
PERSONAL PROTECTION	H

HMIS / NFPA Hazard Rating Legend

* = Chronic Health Hazard	2 = MODERATE
0 = INSIGNIFICANT	3 = HIGH
1 = SLIGHT	4 = EXTREME

Issue date	08-12-2015
Revision date	06-09-2020
Version #	2
List of abbreviations	CAS: Chemical Abstract Service Registration Number ACGIH: American Conference of Governmental Industrial Hygienists TWA: Time Weighted Average STEL: Short Term Exposure Limit LD50: Lethal Dose, 50% LC50: Lethal Concentration, 50% NOEL: No Observed Effect Level COD: Chemical Oxygen Demand BOD: Biochemical Oxygen Demand TOC: Total Organic Carbon IATA: International Air Transport Association IMDG: International Maritime Dangerous Goods Code NFPA: National Fire Protection Association TLV: Threshold Limit Value

TSRN indicates a Trade Secret Registry Number is used in place of the CAS number.

Disclaimer
The information provided in this Safety Data Sheet is correct to the best of our knowledge, information, and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal, and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.